

Mahmoud Elboghday . Jarvis Consulting

I'm a Business Systems Analyst with 3+ years of experience turning operational complexity into measurable business outcomes. Grounded in business fundamentals from a Hospitality Management degree and sharpened through an Information Systems engineering diploma alongside hands-on delivery at Desert Suspension (an authorized 4x4 vehicle equipment dealer operating across Saudi Arabia), where I served as the primary data analyst across Sales, Marketing, and executive leadership, delivering end-to-end data solutions that drove significant improvements in reporting efficiency, inventory accuracy, and profitability.

Skills

Proficient: Python, Machine Learning (scikit-learn, tensorflow, PyTorch), Data Analysis (Pandas, Numpy), SQL/RDBMS (PostgreSQL, SQL Server, Oracle), Power BI/DAX, Agile/Scrum, GCP/AWS, Git/GitHub

Competent: UML/System Design, Docker, ETL Pipeline Development, Linux/Bash, Java/Spring Boot

Familiar: Spark/Databricks, Tableau, Kubernetes, React, UI/UX

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_MahmoudElboghday

Linux Cluster Resource Monitoring System. [GitHub]: Designed and deployed a Linux cluster monitoring agent on Rocky Linux 9 (GCP) to give administrators real-time visibility into CPU, memory, and disk usage across distributed nodes, supporting informed decisions around resource allocation and capacity planning. Built two Bash scripts to collect hardware specs and live usage metrics, persisting data into a centralized PostgreSQL instance containerized via Docker. Automated metric collection with crontab and validated end-to-end pipeline accuracy through SQL-based analysis.

Retail Stock Trading Platform Software Requirements Specification. [GitHub]: Conducted stakeholder elicitation to author a comprehensive Software Requirements Document (SRD) for a full-stack web and mobile retail self-directed trading application commissioned by a Canadian bank. Structured, clear functional boundaries (FRs) managing complex, atomic multi-account portfolios (TFSA, RRSP, Cash Accounts), strict buy/sell trade constraints, and 5-minute interval third-party Market Data API integration flows[cite: 960, 966, 979, 989, 1008]. Engineered robust non-functional requirements targeting high security, system availability, and data integrity parameters for an expected launch volume of 150,000 to 200,000 active clients.

Retail Data Analytics & Customer Segmentation Pipeline. [GitHub]: Developed a Python-based Proof of Concept analytics pipeline inside Docker to convert 1.07 million rows of raw historical transaction data from a PostgreSQL database into actionable marketing intelligence. Built clean, top-down Jupyter Notebook pipelines executing data wrangling, string normalization, snake_case table mappings, and strict split tracking for placed versus cancelled orders to insulate revenue reporting. Engineered custom Recency, Frequency, and Monetary (RFM) features to categorize the customer base into distinct, operational clusters (e.g., Champions, At Risk, Hibernating). Delivered a high-level executive presentation featuring professional Seaborn and Matplotlib visualizations mapping month-over-month growth patterns and demand peaks to guide seasonal stock optimization.

Highlighted Projects

Law Enforcement BI and Resource Optimization Pipeline. [GitHub]: Built an end-to-end crime analytics platform on 1.23M+ Chicago Police Department incidents (2021-2025) using a SQL Server star schema (4 dimension tables, 1 fact table) and a Python ETL pipeline maintaining 96% data completeness across critical fields. Delivered descriptive, diagnostic, predictive, and prescriptive analytics across the full stack. Trained three ML models: a Random Forest crime-type classifier (88% accuracy, top features: hour of day and severity level), a Gradient Boosting volume forecaster (R-squared 0.62, RMSE 50.66 crimes/day, MAPE 21.23%), and a DBSCAN geospatial hotspot detector (epsilon 0.01 approx. 1km radius, min samples 20) that identified 15 high-risk clusters accounting for 35% of all incidents.

Professional Experiences

Machine Learning / Generative AI Developer (Co-op), Sheridan College - Centre for Applied AI (CAAI) (05/2026-present): Translate high-level academic and enterprise research objectives into detailed software specifications, feasibility assessments, and software architecture designs. Design and prototype deep machine learning approaches

utilizing encoder-decoder neural network architectures and context-aware patterns to resolve indoor localization resolution and signal reliability barriers. Evaluate third-party models and open-source frameworks against strict performance requirements to champion data-driven build-versus-buy infrastructure decisions.

Business Analyst Consultant, Jarvis (04/2026-present): Delivering BSA and data projects in an agile, sprint-based environment with structured code reviews and daily standups. Built a Linux cluster monitoring agent using Bash, PostgreSQL, and Docker to automate resource metric collection across a distributed environment.

Business Intelligence Analyst, Desert Suspension (08/2024-03/2026): Owned end-to-end portfolio analytics and operational reporting across three international facilities, serving as the primary liaison between Sales, Marketing, and executive leadership. Automated ETL workflows using Python and SQL, reducing manual reporting cycles by 75% and eliminating cross-departmental data silos. Applied statistical trend analysis to inventory data, achieving 85% forecasting accuracy and contributing to a 17% increase in net profit. Governed a centralized data warehouse maintaining integrity and usability standards across all pipelines. Produced audit-ready technical and business documentation ensuring reproducibility and knowledge continuity across reporting deliverables. Operated in an agile workflow, partnering with cross-functional stakeholders to align data outputs with strategic business priorities.

Data Analyst, Upwork (02/2023-01/2024): Delivered freelance data analytics engagements across multiple clients, producing over 10 Power BI dashboards with real-time KPI tracking that accelerated decision-making speed by 40% for 15+ stakeholders. Built automated SQL ETL pipelines consolidating multiple data sources at 98% accuracy, and designed A/B tests using Python statistical methods that drove a 15% improvement in customer engagement metrics.

Education

Sheridan College (2025-present), Advanced Diploma, Information Systems Engineering

University of Alexandria (2015-2019), Bachelor of Science, Hotel Management and Hospitality

Miscellaneous

- IBM Data Science Professional Certificate
- DataCamp Power BI Data Analyst Certificate
- PwC Data Analytics Job Simulation
- Accenture Data Analytics Job Simulation
- Competitive athlete across multiple disciplines: started as a competitive swimmer, progressed to men's physique bodybuilding, and competed in CrossFit team events across the Middle East (Saudi Arabia and Bahrain).
- Currently focused on staying active and healthy as a new father, training to keep up with my daughter as she grows.
- Content creator with over 300k followers covering Canadian immigration topics, license process for foreign-trained professionals by interviewing Licensed fellows who went through the journey and are now ready to pay it back to the community.